

Computer Graphics

CSIT — 3rd Semester — 2081 — Answer Sheet
Subject Code: CSC214

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Derive the expression for Bresenham's Line Drawing algorithm. Trace the points in the line path with starting point (6, 12) and end point (10, 5) using Bresenham's line drawing algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is the major drawback of Sutherland Hodgeman Polygon Clipping Algorithm? Illustrate with suitable example. Explain B-spline curve and its properties.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain RGB color model. Differentiate between virtual reality and augmented reality with examples. Explain Binary Space Partitioning (BSP) trees with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. What do you mean by refresh rate of a display device? Explain vector scan display with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Plot the first octant of a circle centered at (-2, -2), having radius 5 units using mid-point circle algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Given a triangle with vertices A(2,3), B(5,5), C(4,3) by rotating 90 degree about the origin and then translating two unit in each direction. Use homogenous transformation matrix to find the new vertices of the triangle.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Construct the Bezier curve with the following polygon vertices (control points): A(1,1), B(2,3), C(4,3), and D(6,4).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Compare Scanline Method for Visible Surface Detection with Depth Buffer Method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain any two functions in OpenGL. Write the basic commands to draw the pixel and polygon in OpenGL.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Differentiate between parallel projection and Perspective projections.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Describe about any two basic illumination models.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on :a) Sweep Representationb) Intensity Attenuation

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.